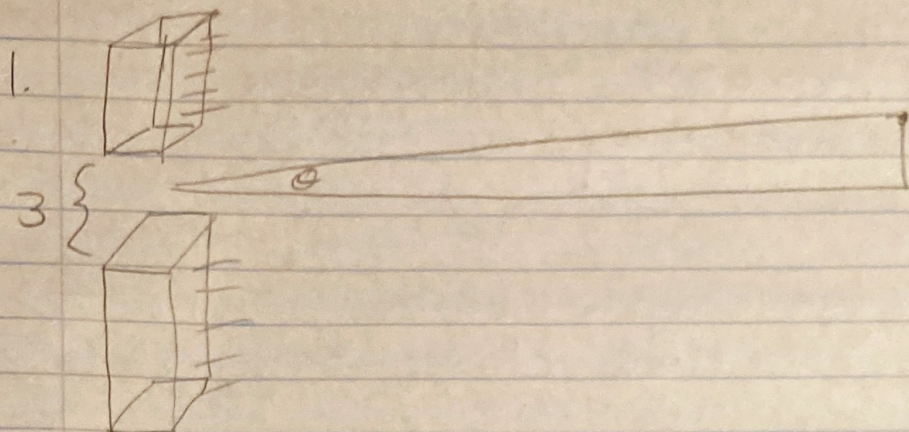


WPS 23

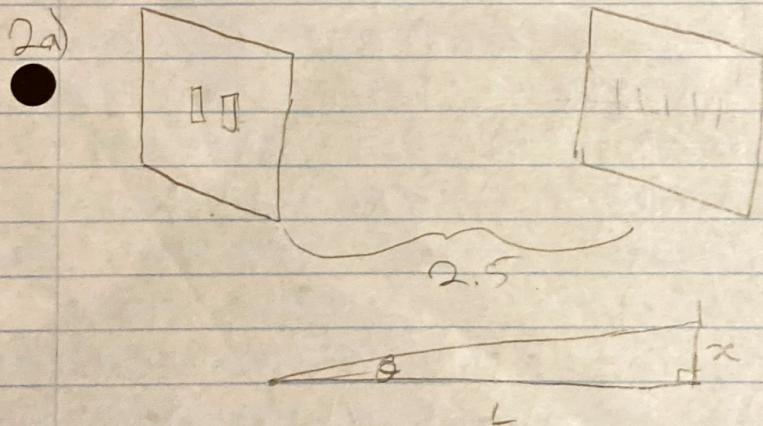
CRSALE010



• We need that $d \sin \theta = \frac{\lambda}{2}$ for $d=3$
 Now $T = \frac{1}{f}$, so $\lambda = \frac{v}{f}$ $VT = \frac{v}{f}$ for speed v .
 The speed of sound $\approx 343 \text{ m s}^{-1}$, so
 $\lambda = \frac{343}{440} = 0.780 \text{ m}$

So we need $3 \sin \theta = \frac{0.780}{2}$
 $\Rightarrow \theta = \arcsin\left(\frac{0.78}{2 \cdot 3}\right) = 7.47^\circ$

So we need to be on the line that makes 7.47° with the horizontal.



$$\sin \theta \approx \tan \theta = \frac{x}{L}$$

$$\text{Also } d \sin \theta = \lambda \Rightarrow \frac{\lambda}{\theta} = \frac{x}{L}$$

$$\Rightarrow x = \frac{L \lambda}{\theta}$$

$$= \frac{2.5 \times 550 \times 10^{-3}}{0.1 \times 10^{-3}} = 0.0138 \text{ m}$$

b) $x = \frac{\lambda L}{d}$

So if d increases, x decreases so they get closer together

c) $x = \frac{\lambda L}{d}$

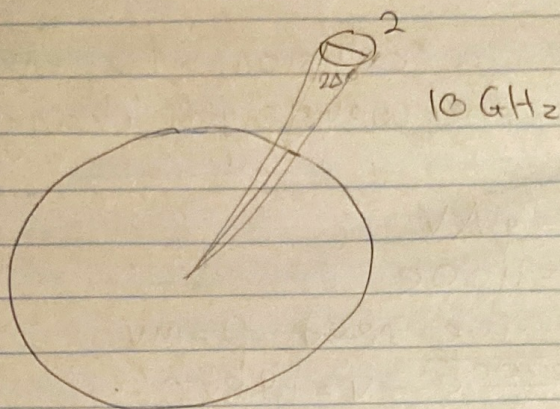
A smaller wavelength creates smaller x , thus the lines are closer together

3a) A low-frequency mode with a long wavelength is easier to sustain than a higher frequency mode and the higher frequency mode will dampen out quicker

b) Pressing down on the string creates a node at that point. Normally the allowed wavelengths of a string are in the form $\frac{2L}{n}$, $n \in \mathbb{N}$, (such that the waves are 0 at the ends). But if we have a node in the midpoint, then we need $\lambda = \frac{L}{n}$, $n \in \mathbb{N}$. ^{since $\frac{2L}{n} = \frac{L}{n/2}$} (2L will not have a node at the midpoint). The odd harmonics are thus not possible.

4. If only blue and red light is present, then green light is absent. There thus must be 0 re-radiation. This is possible if the thickness is any integer multiple of $\frac{\lambda}{2}$. Now $\lambda_{\text{green}} = 550 \text{ nm}$ (Q2), so $\lambda' = \frac{\lambda_{\text{green}}}{1.3}$
 $= \frac{550}{1.3} = 423 \text{ nm}$, thus one possible answer is thickness $= \frac{423}{2} = 212 \text{ nm}$

5.



$$2\Delta\theta = 2 \cdot \frac{\lambda}{l}$$

$$\text{Now } \lambda f = v$$

$$\lambda = \frac{v}{f}$$

Taking $v=c$ (radio waves propagate at speed of light)

$$\text{Then } \lambda = \frac{c}{f} = \frac{3 \times 10^8}{10 \times 10^9} = 0.03 \text{ m}$$

$$\text{So } 2\Delta\theta = 2 \cdot \frac{0.03}{l} = 0.03 \text{ rad}$$

$$\text{Now } l = r\theta$$

$$= r_e \theta$$

$$= 6.4 \times 10^6 \cdot 0.03$$

$$= 1.92 \times 10^5 \text{ m}$$

6. The intensity I is the power per unit area. So $I = \frac{P}{A} = \frac{100}{\pi(0.1)^2} = 3183$.

Now $I = \text{energy flux}$ so $I = \frac{1}{\mu_0} E B$

$$= \frac{1}{\mu_0} E \cdot \frac{E}{c}$$

$$\text{So } E = \sqrt{\mu_0 c I}$$

$$= 1095 \text{ N/C}$$

For a power of 100W, 100J is transferred in 1 second. Now $E_{\text{photon}} = \frac{hc}{\lambda} \approx \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{550 \times 10^{-9}}$

$$\approx 3.6 \times 10^{-19} \text{ J}$$

Thus there are $\frac{100}{3.6 \times 10^{-19}} = 2.78 \times 10^{20}$ photons

7. $E = \frac{hc}{\lambda}$. Now after a collision, the energy decreases and so the wavelength changes

8. We know $\Delta K = q\Delta V$
 $= 1100e$

Now using $\lambda = \frac{h}{p}$ we need $p = mv$

$$\frac{1}{2}mv^2 = 1100e \Rightarrow \sqrt{\frac{2 \cdot 1100e}{m_e}} = v = 1.98 \times 10^7$$

$$\text{So } p = mv = 1.78 \times 10^{-23}$$

$$\text{So } \lambda = 3.71 \times 10^{-11}$$

For a ring, we need constructive interference. So $2d \sin \theta = 2 \times 10^{-10} \sin \theta = \lambda$
 $\theta = 10.7^\circ$